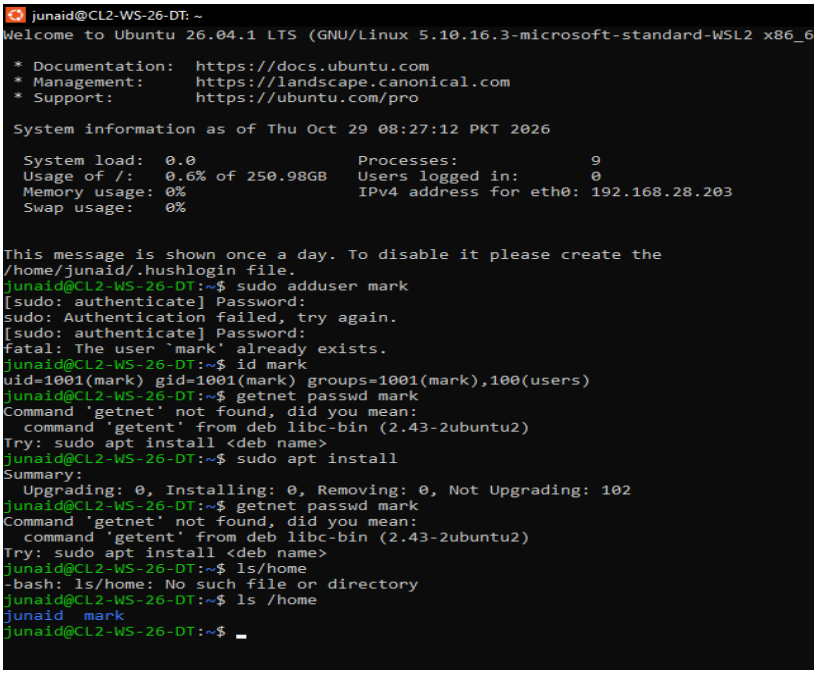
Linux User and Group Management

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Question 1

Create user mark and verify

```
sudo adduser mark
id mark
getent passwd mark
ls /home
```



Explanation: adduser is an interactive, friendly script. It creates the user, a home directory (/home/mark), a private group with the same name, copies default files from /etc/skel and asks for a password and optional details (full name, phone etc.). The id command shows the UID, GID and groups, and getent passwd mark shows the entry, so together they prove the account exists.

Question 2

Set the password for mark

```
sudo passwd mark
```

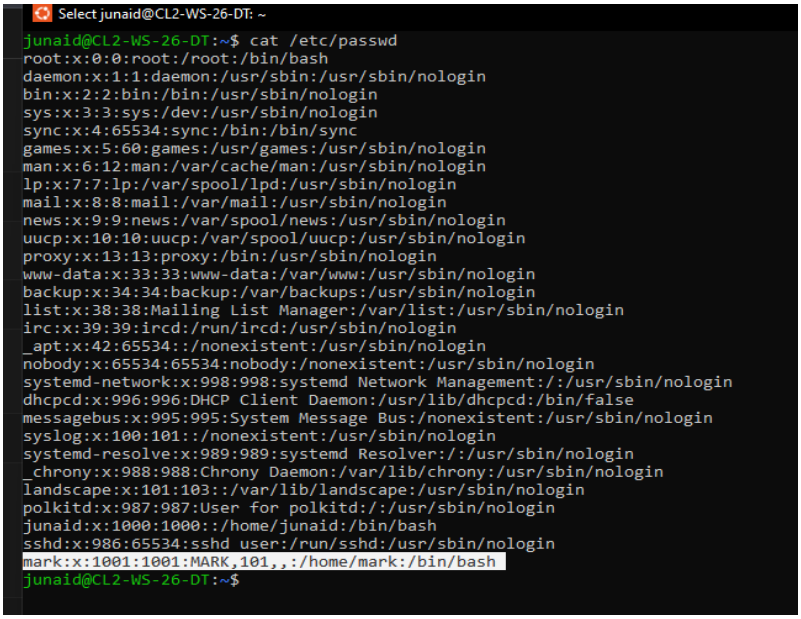
```
junaid@CL2-WS-26-DT: ~
junaid@CL2-WS-26-DT:~$ ls /home
junaid  mark
junaid@CL2-WS-26-DT:~$ sudo passwd mark
New password:
Retype new password:
passwd: password updated successfully
junaid@CL2-WS-26-DT:~$
```

Explanation: passwd sets a new password; you type it twice and nothing is shown on the screen while typing. Linux requires a password for authentication: it proves that the person logging in is really the owner of the account. Without one, anyone could log in as that user and read or change the user's files. The password is stored only as a salted hash in /etc/shadow, never as plain text.

Question 3

Display /etc/passwd and explain the mark entry

```
cat /etc/passwd
grep '^mark:' /etc/passwd
```



A typical entry looks like: mark:x:1001:1001:,,,:/home/mark:/bin/bash

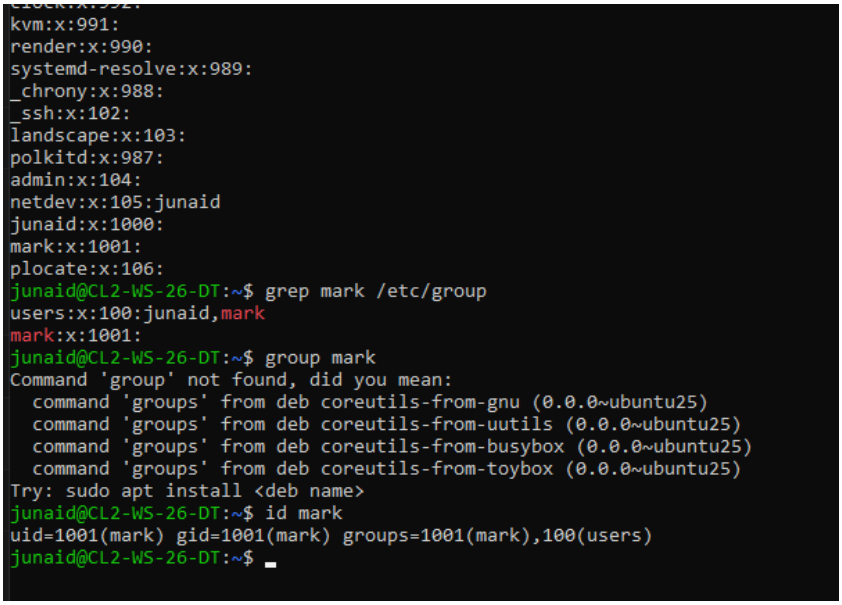
Field	Example	Meaning
1 - Username	mark	Login name of the account.
2 - Password	x	The 'x' means the real password hash is stored in /etc/shadow.
3 - UID	1001	User ID, the number the kernel uses to identify the user.
4 - GID	1001	Primary group ID of the user (group 'mark' in /etc/group).

5 - GECOS / comment	(empty or ,,,)	Full name or description of the user.
6 - Home directory	/home/mark	Directory the user lands in after login.
7 - Login shell	/bin/bash	Program started when the user logs in.

Question 4

Display /etc/group and find mark's groups

```
cat /etc/group
grep mark /etc/group
groups mark
id mark
```



Explanation: /etc/group lists every group in the format groupname:x:GID:member-list. System groups such as root, sudo, adm, users, www-data, shadow and many others exist by default. mark has his own primary group 'mark' (GID 1001) and, on many Ubuntu systems, also appears in the 'users' group. The command groups mark or id mark shows exactly which groups mark belongs to.

Question 5

Display /etc/hosts

```
cat /etc/hosts
```

```
junaid@CL2-WS-26-DT:~$ cat /etc/hosts
# This file was automatically generated by WSL. To stop automatic generation of this file, add the following entry to /e
tc/wsl.conf:
# [network]
# generateHosts = false
127.0.0.1       localhost
127.0.1.1       CL2-WS-26-DT.SHU.NET    CL2-WS-26-DT

172.16.23.34    host.docker.internal
172.16.23.34    gateway.docker.internal
127.0.0.1       kubernetes.docker.internal

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
junaid@CL2-WS-26-DT:~$
```

Explanation: /etc/hosts is a local text file that maps hostnames to IP addresses. The system checks it before asking a DNS server, so it is useful for quick local name resolution and testing. The localhost entries are '127.0.0.1 localhost' (IPv4 loopback) and '::1 localhost ip6-localhost ip6-loopback' (IPv6 loopback). There is usually also a line such as '127.0.1.1 <your-hostname>'. These addresses point back to the same computer.

Question 6

```
junaid@CL2-WS-26-DT: ~
junaid@CL2-WS-26-DT:~$ sudo cat /etc/shadow | cut -d: -f1,3-9
root:20692:0:99999:7:::
daemon:20692:0:99999:7:::
bin:20692:0:99999:7:::
sys:20692:0:99999:7:::
sync:20692:0:99999:7:::
games:20692:0:99999:7:::
man:20692:0:99999:7:::
lp:20692:0:99999:7:::
mail:20692:0:99999:7:::
news:20692:0:99999:7:::
uucp:20692:0:99999:7:::
proxy:20692:0:99999:7:::
www-data:20692:0:99999:7:::
backup:20692:0:99999:7:::
list:20692:0:99999:7:::
irc:20692:0:99999:7:::
_apt:20692:0:99999:7:::
nobody:20692:0:99999:7:::
systemd-network:20692::::1:
dhcpcd:20692::::1:
messagebus:20692:::::
syslog:20692:::::
systemd-resolve:20692::::1:
_chrony:20692:::::
landscape:20692:::::
polkitd:20692:::::
junaid:20735:0:99999:7:::
sshd:20738:::::
mark:20755:0:99999:7:::
junaid@CL2-WS-26-DT:~$
```

Inspect /etc/shadow (do NOT show hashes)

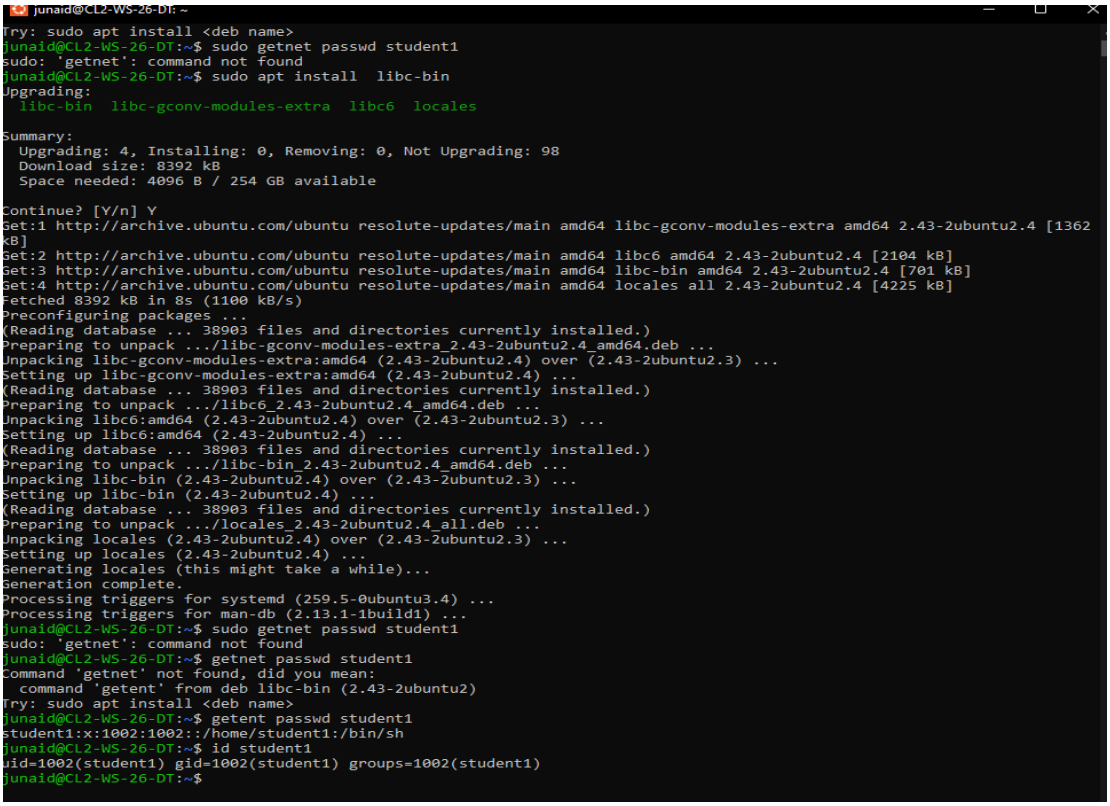
```
sudo cat /etc/shadow | cut -d: -f1,3-9
ls -l /etc/shadow
```

Explanation: The command above uses cut to hide field 2 (the hash) so your screenshot is safe. /etc/shadow stores each user's encrypted (hashed) password, the date of the last password change, minimum and maximum password age, warning days, inactivity period and account expiry date. Access is restricted (permissions -rw-r----- root:shadow) because anyone who could read the hashes could try to crack passwords offline with brute-force or dictionary attacks. That is why /etc/passwd is readable by everyone but the hashes live in /etc/shadow, which only root can read.

Question 7

Create student1 with useradd

```
sudo useradd student1
sudo passwd student1
getent passwd student1
id student1
```

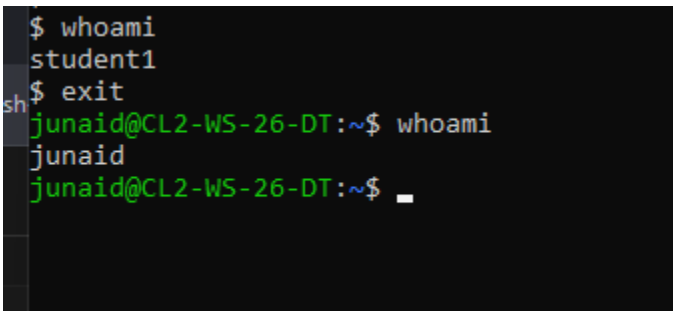


Explanation: useradd is the low-level command. By default it does not create a home directory, does not ask for a password and usually assigns /bin/sh as the shell, so you must set these yourself (for example useradd -m -s /bin/bash student1). adduser is a friendlier Debian/Ubuntu wrapper script around useradd: it is interactive, creates the home directory automatically, copies /etc/skel files and prompts for a password. That is the main difference.

Question 8

Switch to student1 using su

```
su student1
whoami
exit
whoami
```

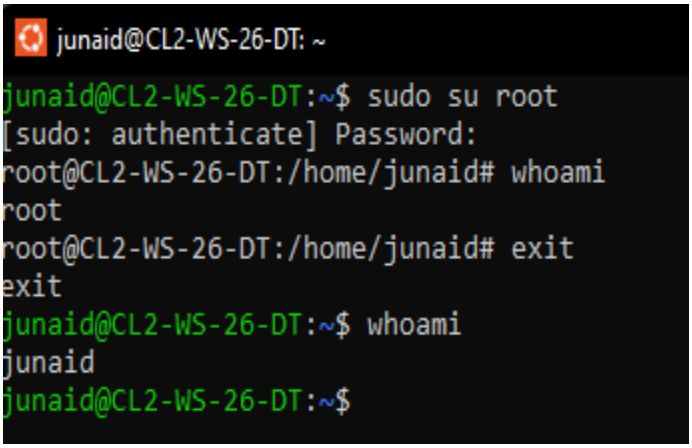


Explanation: su (substitute user / switch user) starts a new shell as another user. Because student1 has no home directory, you may see a warning such as 'No directory, logging in with HOME=/' which is normal here. whoami prints student1 while inside the new shell. exit returns you to your original account (junaid), which the second whoami confirms. Use 'su - student1' for a full login shell with that user's environment.

Question 9

Switch to root

```
sudo su root
whoami
exit
whoami
```



Explanation: sudo su root gives you a root shell after you enter your own password; the prompt changes from $ to #, and whoami shows root. exit brings you back. Root has unlimited power: it can read, change or delete any file, add or remove users, stop services and even destroy the operating system with one wrong command (for example a mistaken rm -rf). There is no confirmation or undo, so root should be used only when necessary, for the shortest time possible. Normal work should be done as a regular user with sudo for single commands.

Question 10

Set the comment field

```
sudo usermod -c "MARK" mark
getent passwd mark
```



Explanation: -c changes the comment (GECOS) field, which is the 5th field of /etc/passwd. After the command the entry becomes mark:x:1001:1001:MARK:/home/mark:/bin/bash. This field normally stores the user's full name or description.

Question 11

Change UID of mark to 1003

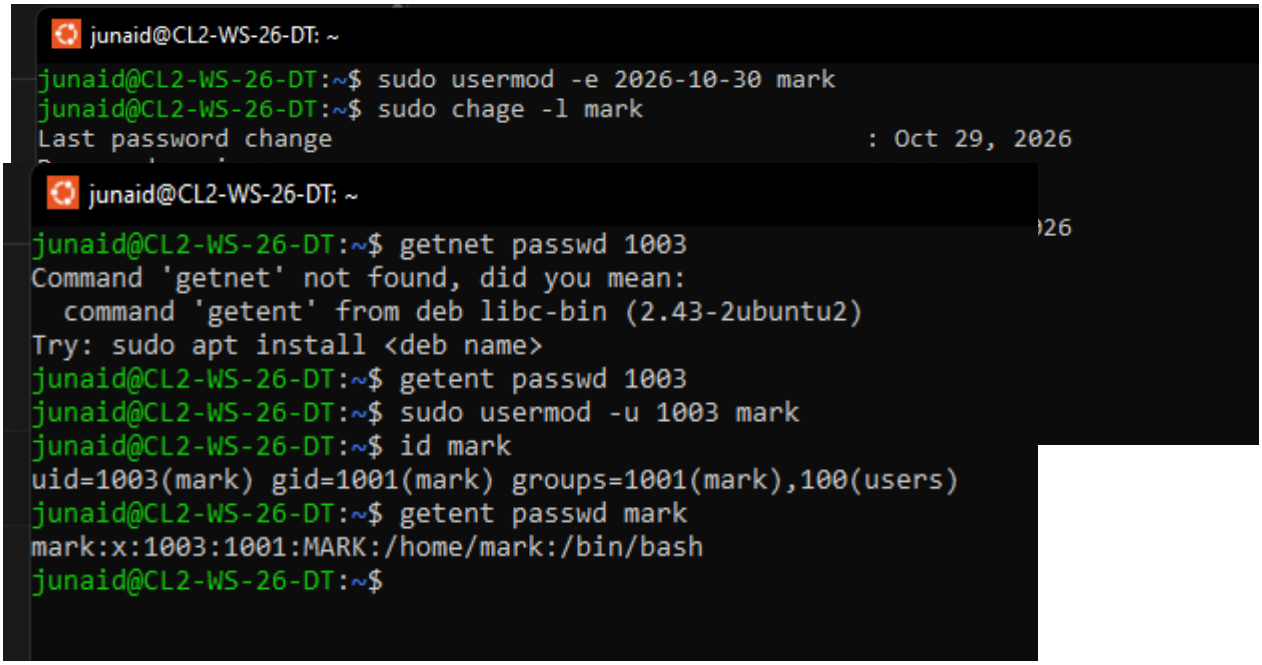
```
getent passwd 1003
sudo usermod -u 1003 mark
id mark
getent passwd mark
```

Explanation: First check that UID 1003 is free: if getent passwd 1003 prints nothing, it is available (if it prints a line, choose another number). Then usermod -u changes the UID and also updates ownership of files in mark's home directory. A UID (User Identifier) is the unique number the Linux kernel uses to identify a user; usernames are only human-friendly labels mapped to UIDs. File ownership and permissions are all checked using the UID, and UID 0 is always root.

Question 12

Set an account expiration date

```
sudo usermod -e 2026-10-30 mark
sudo chage -l mark
```

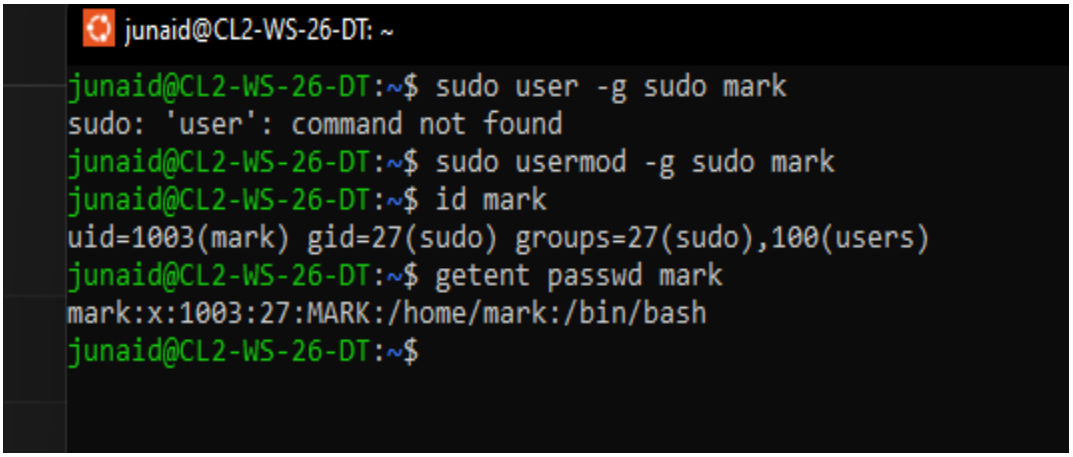


Explanation: -e sets the date on which the account becomes disabled. chage -l lists the ageing information, and the line 'Account expires' should show Oct 30, 2026. After that date the user cannot log in anymore (login is rejected), but the account, home directory and files remain and the administrator can re-enable it with sudo usermod -e '' mark. It is useful for temporary users such as interns or contractors.

Question 13

Change primary group to sudo

```
sudo usermod -g sudo mark
id mark
getent passwd mark
```



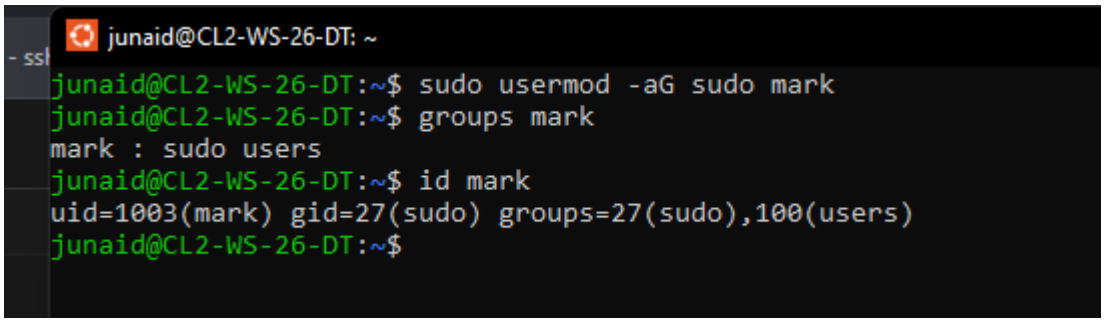
Explanation: -g replaces the primary group of mark with sudo (GID 27), so the 4th field in /etc/passwd becomes 27 and id mark shows gid=27(sudo). The primary group is the group assigned to new files and directories created by the user, and every user has exactly one. Supplementary groups are additional groups that give extra permissions (for example access to shared folders or devices); a user can belong

to many of them. Note: giving a normal user the sudo group also gives administrator rights, so do this only in the lab.

Question 14

Add mark to supplementary group sudo

```
sudo usermod -aG sudo mark
groups mark
id mark
```

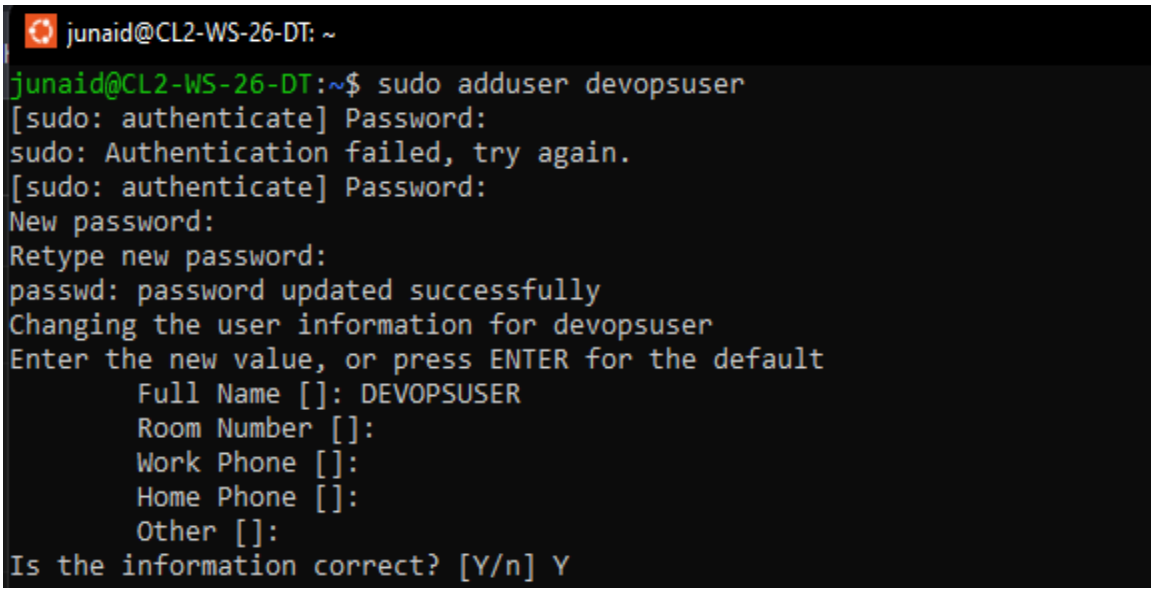


Explanation: -G sets the supplementary group list, and -a means append. With -aG, mark is added to sudo while keeping all his existing supplementary groups. Without -a, usermod -G sudo mark would replace the whole list, silently removing mark from every other supplementary group and possibly breaking his access to files, devices or services. That is why -aG is the safer choice.

Question 15 - Practical Challenge: User Lifecycle Management

Step 1 - Create the user

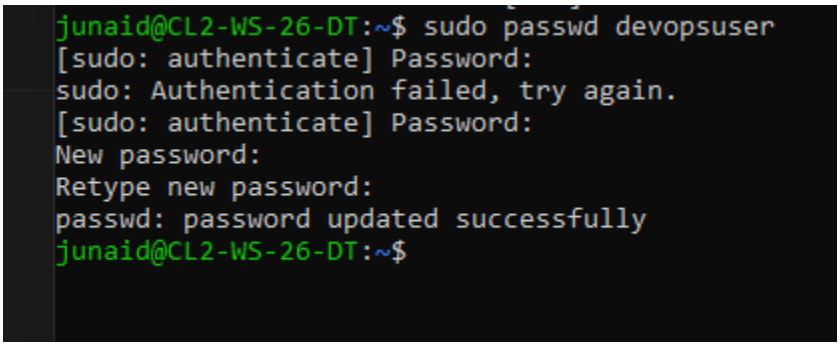
```
sudo adduser devopsuser
```



Explanation: adduser creates the account, home directory /home/devopsuser and a private group.

Step 2 - Set the password

```
sudo passwd devopsuser
```



Explanation: Sets the login password (stored as a hash in /etc/shadow).

Step 3 - Verify entry and groups

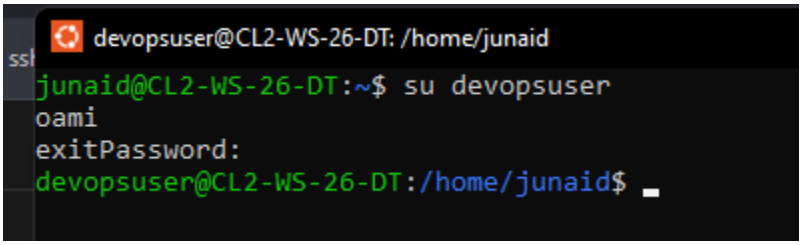
```
grep '^devopsuser:' /etc/passwd
id devopsuser
groups devopsuser
```



Explanation: Shows the passwd entry with its UID/GID, and that the user belongs to the primary group devopsuser.

Step 4 - Switch user and verify

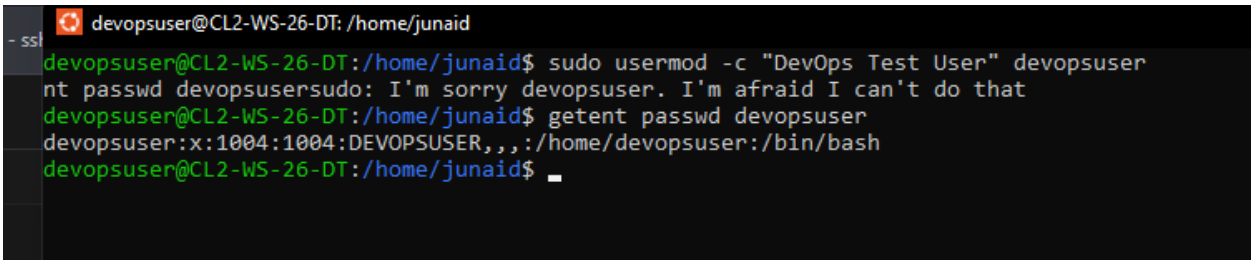
```
su devopsuser
whoami
exit
```

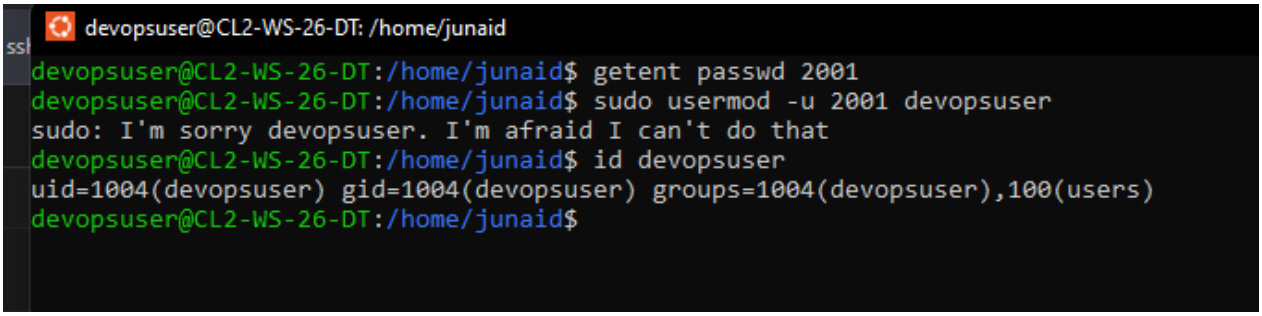


Explanation: whoami prints devopsuser, proving the shell now runs as the new user; exit returns to your own account.

Step 5 - Update comment field

```
sudo usermod -c "DevOps Test User" devopsuser
getent passwd devopsuser
```





Explanation: The 5th field of /etc/passwd now shows the new description.

Step 6 - Change UID to an available one

```
getent passwd 2001
sudo usermod -u 2001 devopsuser
id devopsuser
```

Explanation: Empty output of the first command shows 2001 is free; id then shows uid=2001.

Step 7 - Set expiration date

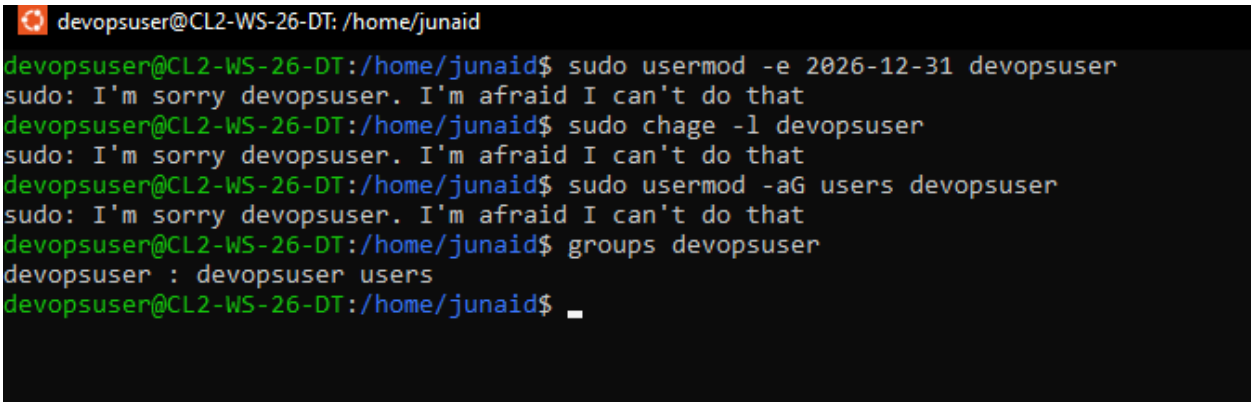
```
sudo usermod -e 2026-12-31 devopsuser
sudo chage -l devopsuser
```

[Paste your own terminal screenshot here - Q15 / Step 7]

Explanation: chage -l shows 'Account expires : Dec 31, 2026'.

Step 8 - Add to a supplementary group

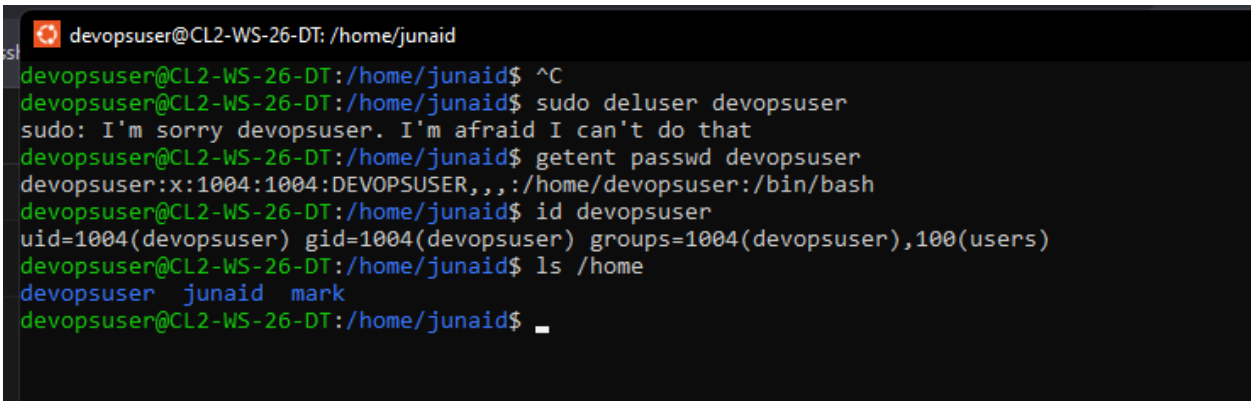
```
sudo usermod -aG users devopsuser
groups devopsuser
```



Explanation: The user is appended to the 'users' group (use a harmless group such as users or developers; avoid sudo for a test user) and keeps existing groups.

Step 9 - Delete the user and verify

```
sudo deluser devopsuser
# or: sudo userdel devopsuser
getent passwd devopsuser
id devopsuser
ls /home
```



Explanation: The account is removed from /etc/passwd, so getent prints nothing and id says 'no such user'. The home directory stays unless you use sudo deluser --remove-home devopsuser or userdel -r.

Summary of the lifecycle: create -> set password -> verify -> switch -> modify (comment, UID, expiry, groups) -> delete. Each stage changed a different part of /etc/passwd, /etc/shadow or /etc/group, and the verification commands (id, getent, chage -l, groups) confirm every change.